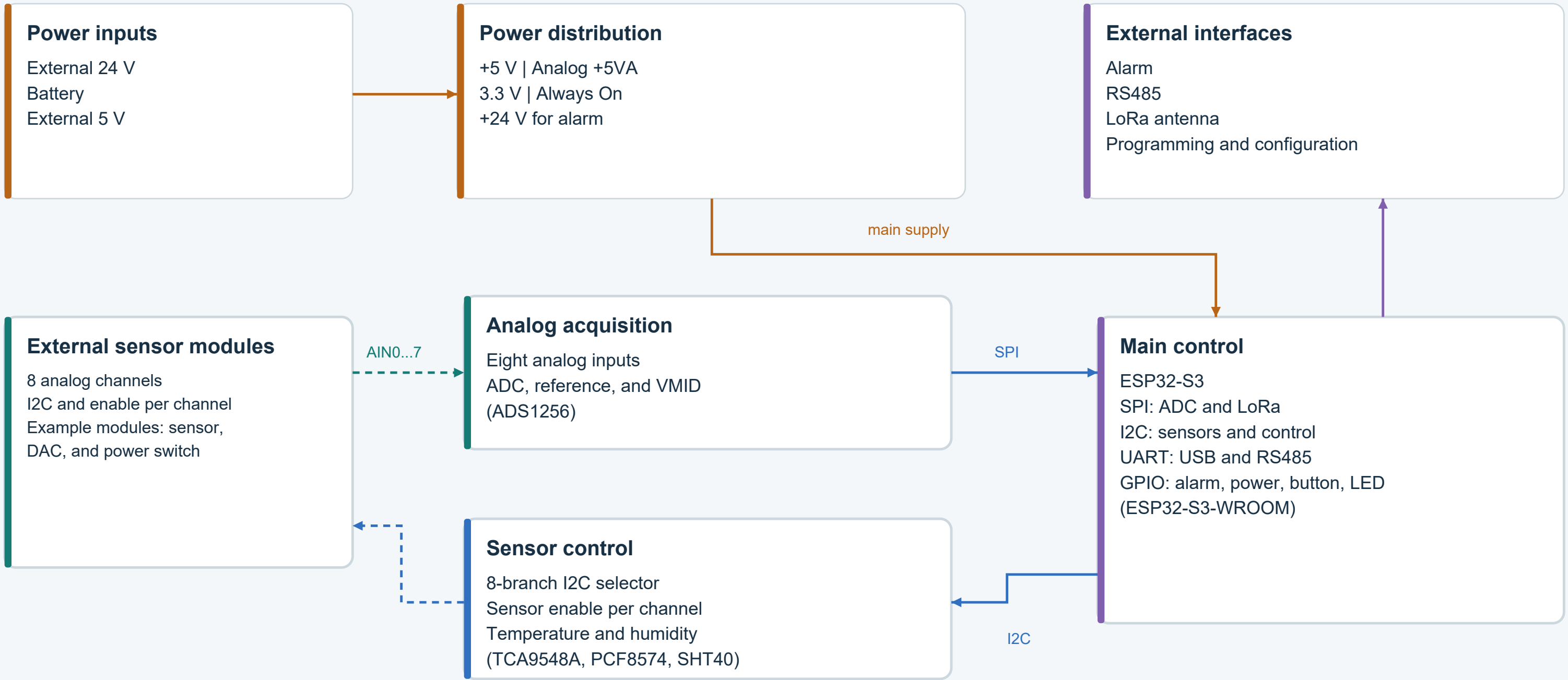


01 System architecture

Power, sensor acquisition, control, and communications in one device.

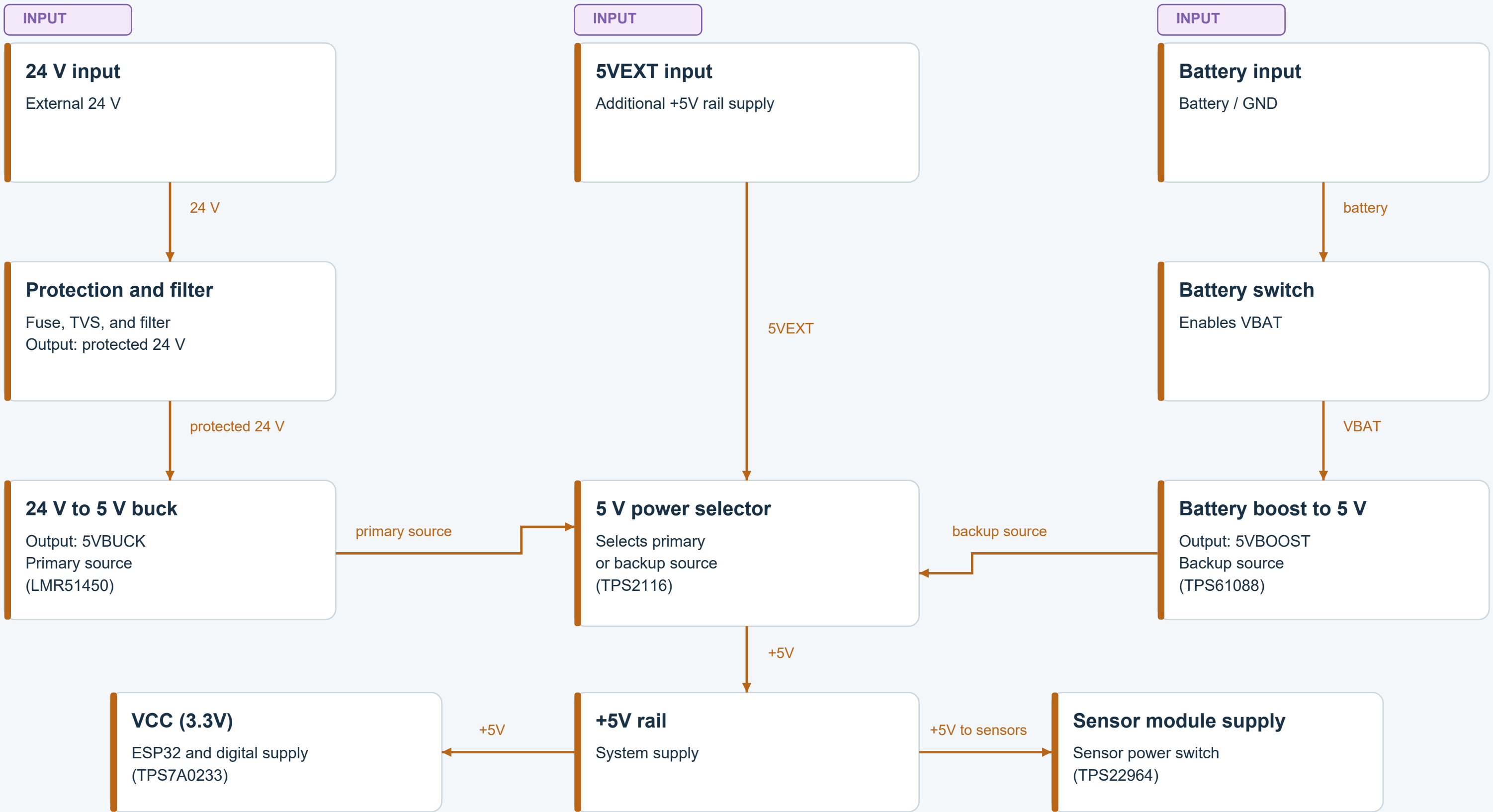


How to read

Brown: power. Green: analog. Blue: digital communication. Purple: control and status.
Dashed lines show interconnection between sensor modules.

02 Main power distribution

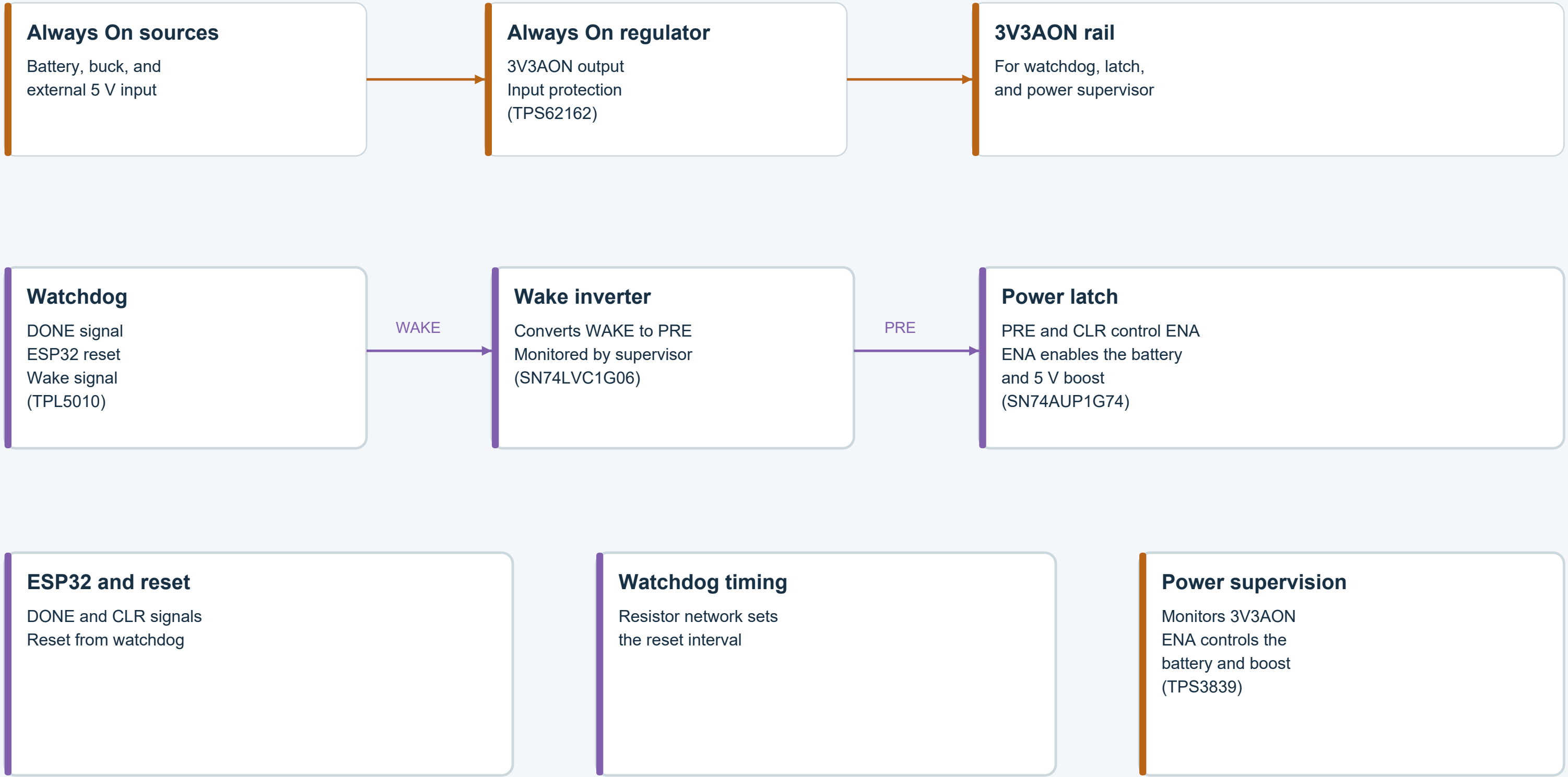
Three power inputs, 5 V source selection, analog rail, and sensor supply.



Protected 24 V: output after the fuse, TVS, and filter.

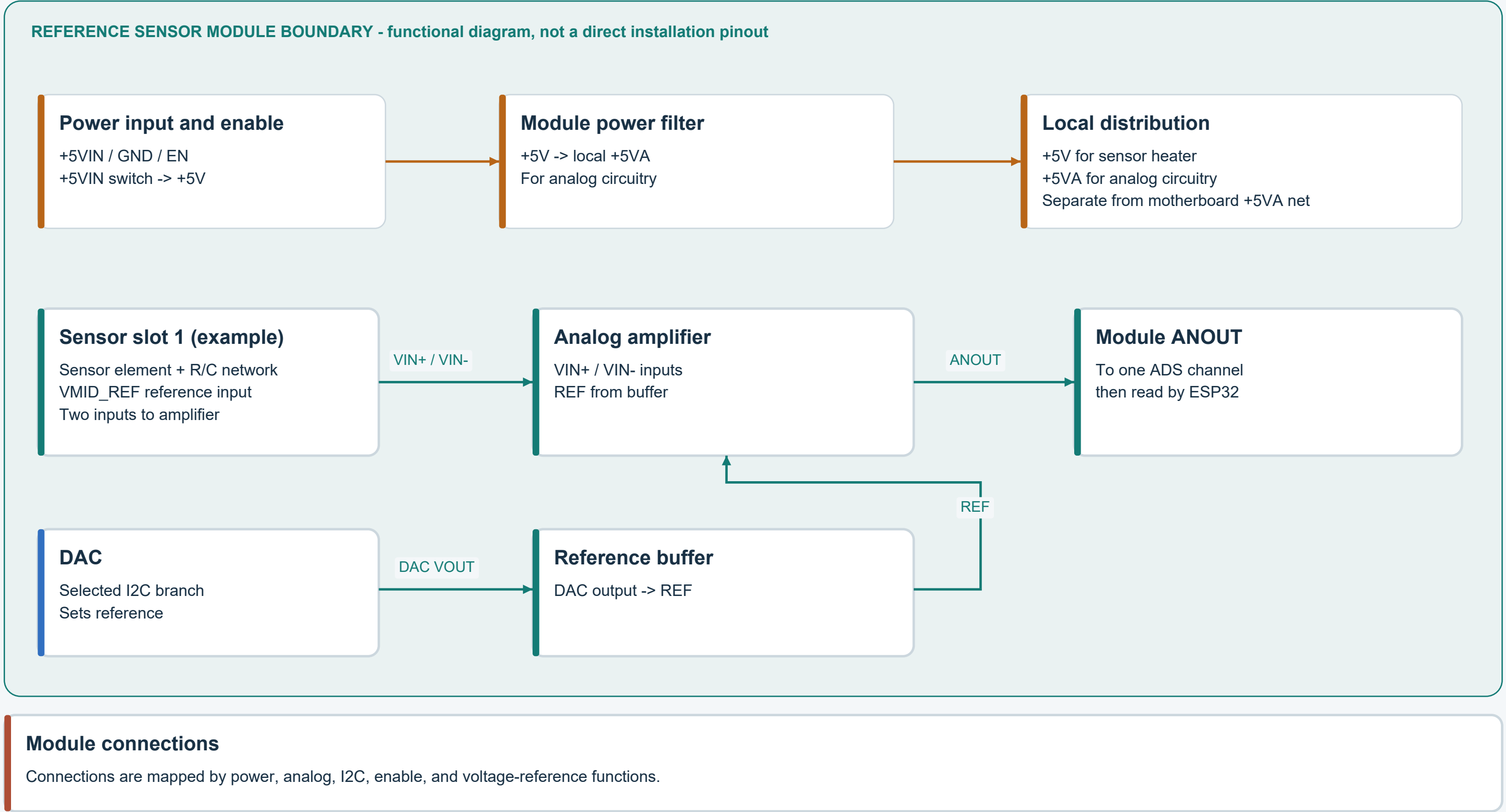
03 Always On, watchdog, and latch

Always On (AON) is a continuously active power rail for supervision and power control.



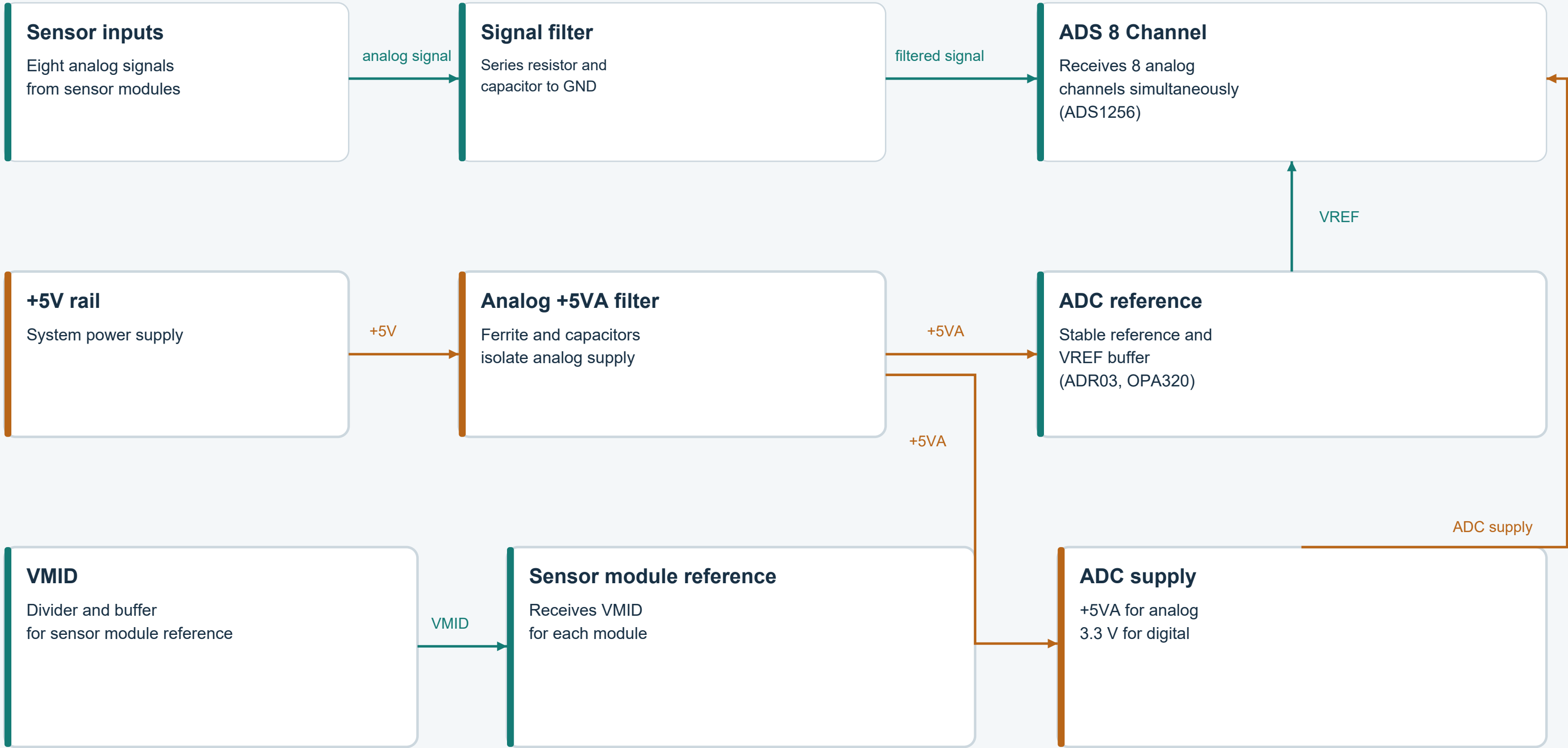
04 Sensor module: functional path

Each module receives power, I2C, enable, and provides one analog output.



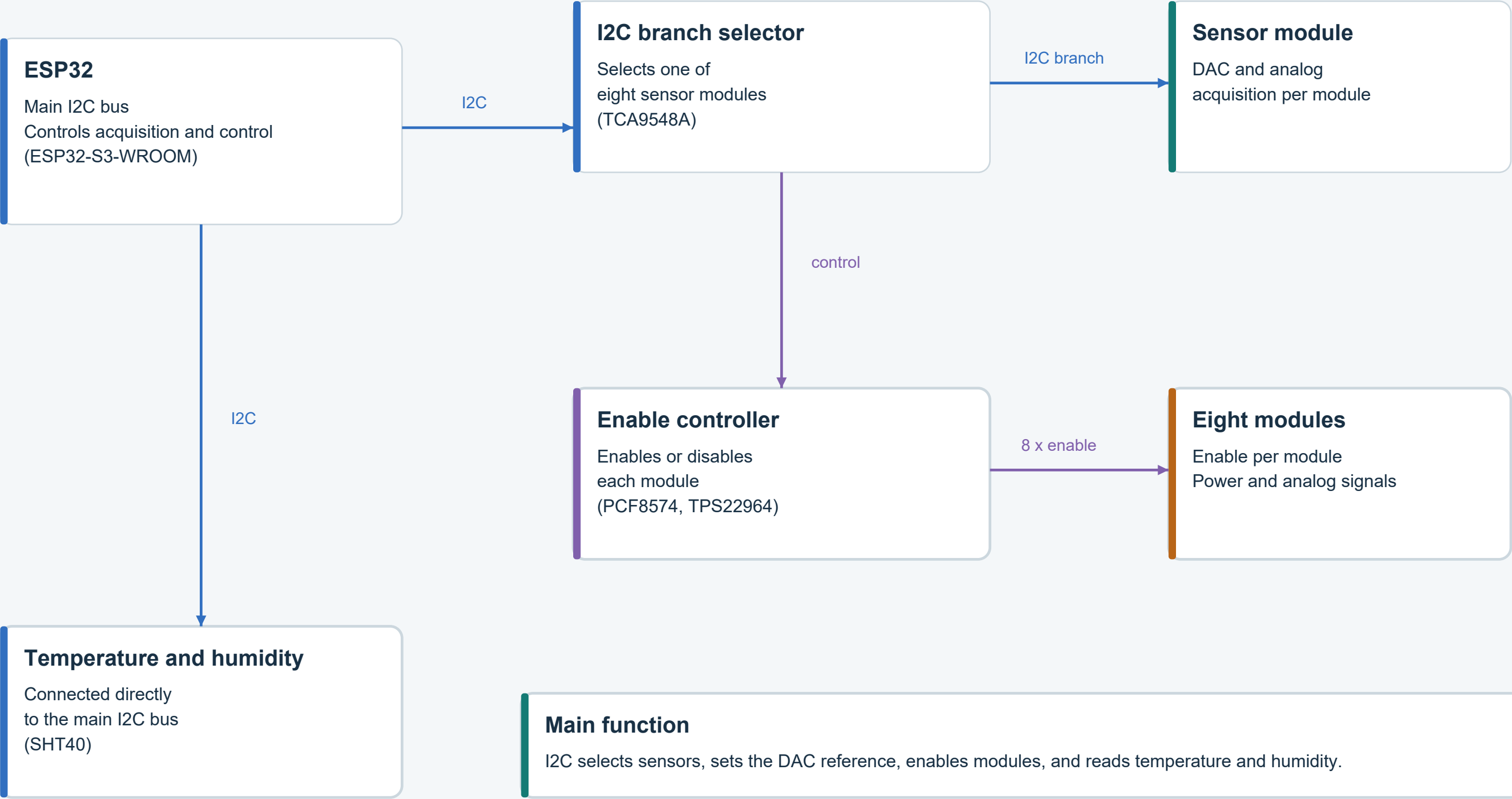
05 Analog acquisition and voltage reference

The analog path has a separate supply filter, voltage reference, VMID, and ADS 8 Channel.



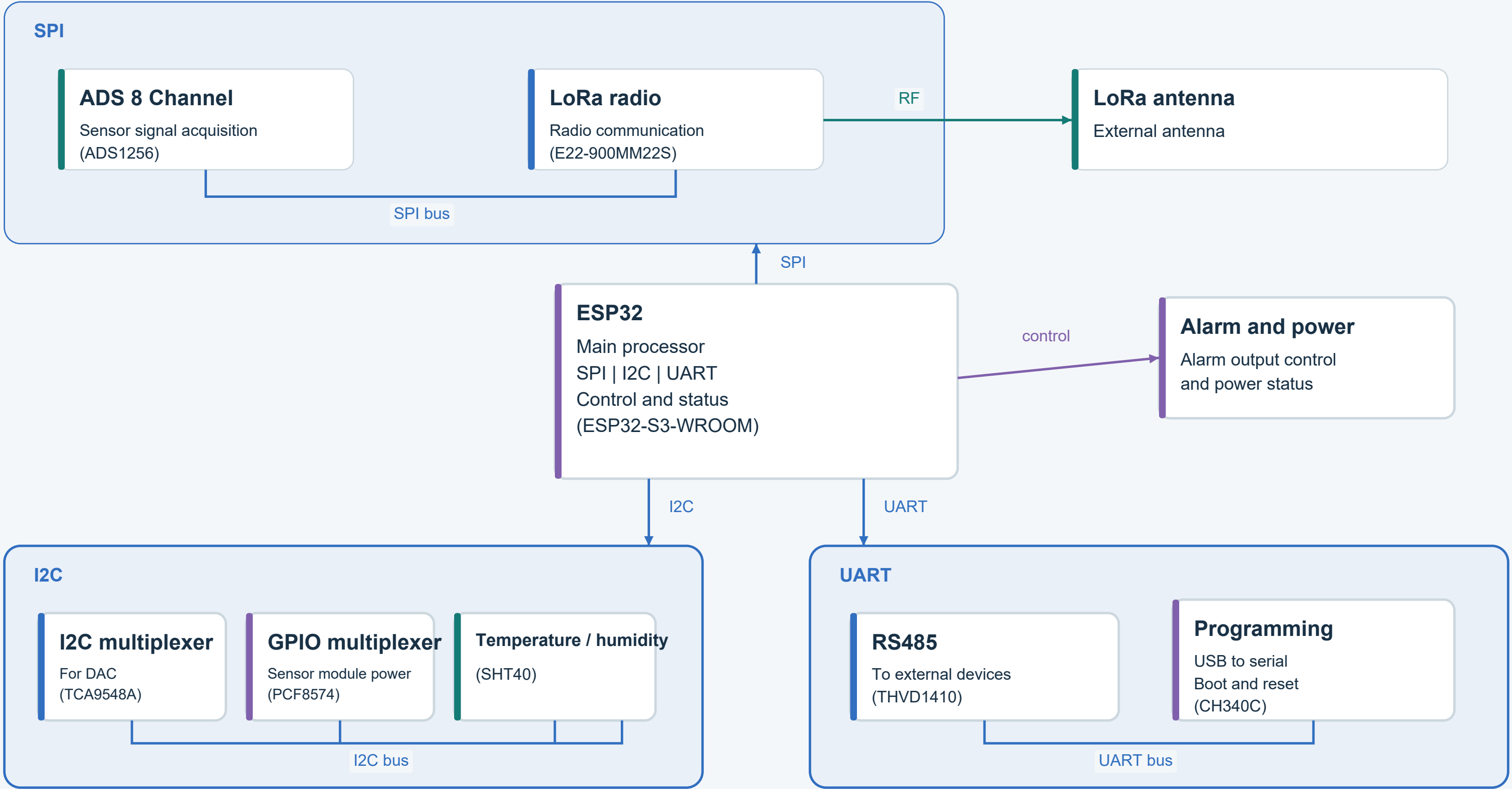
06 I2C and sensor control

I2C selects sensor branches, sets the DAC, and controls each module power supply.



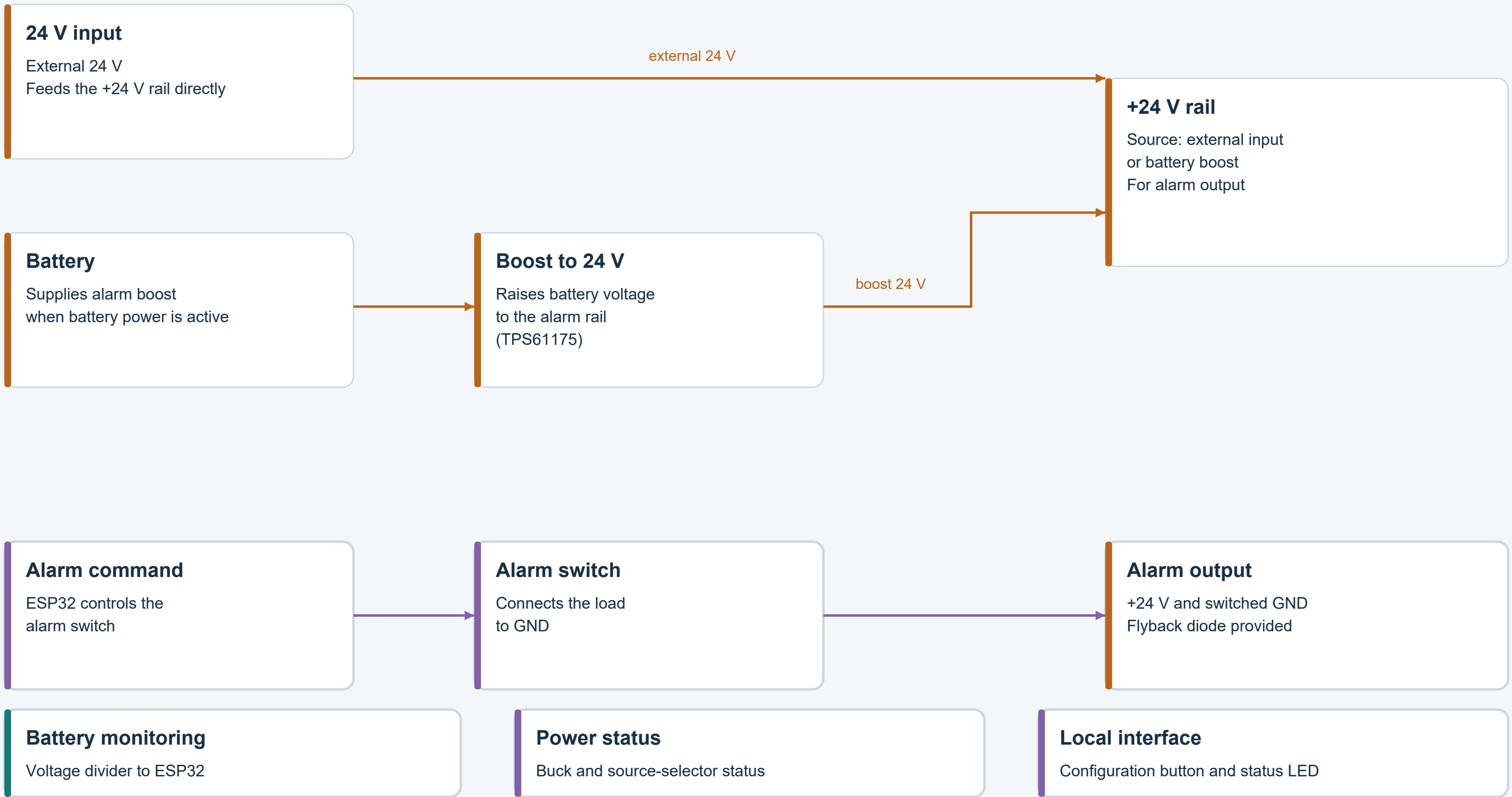
07 ESP32 communications and connections

Connections are grouped by SPI, I2C, and UART buses; device control is separate from communications.



08 Alarm, battery, and power status

The +24 V alarm rail can be supplied by external 24 V input or battery boost.



09 Energy, data, and control flow

Power supplies devices; sensor data flows to the ESP32; the ESP32 controls communications and outputs.

